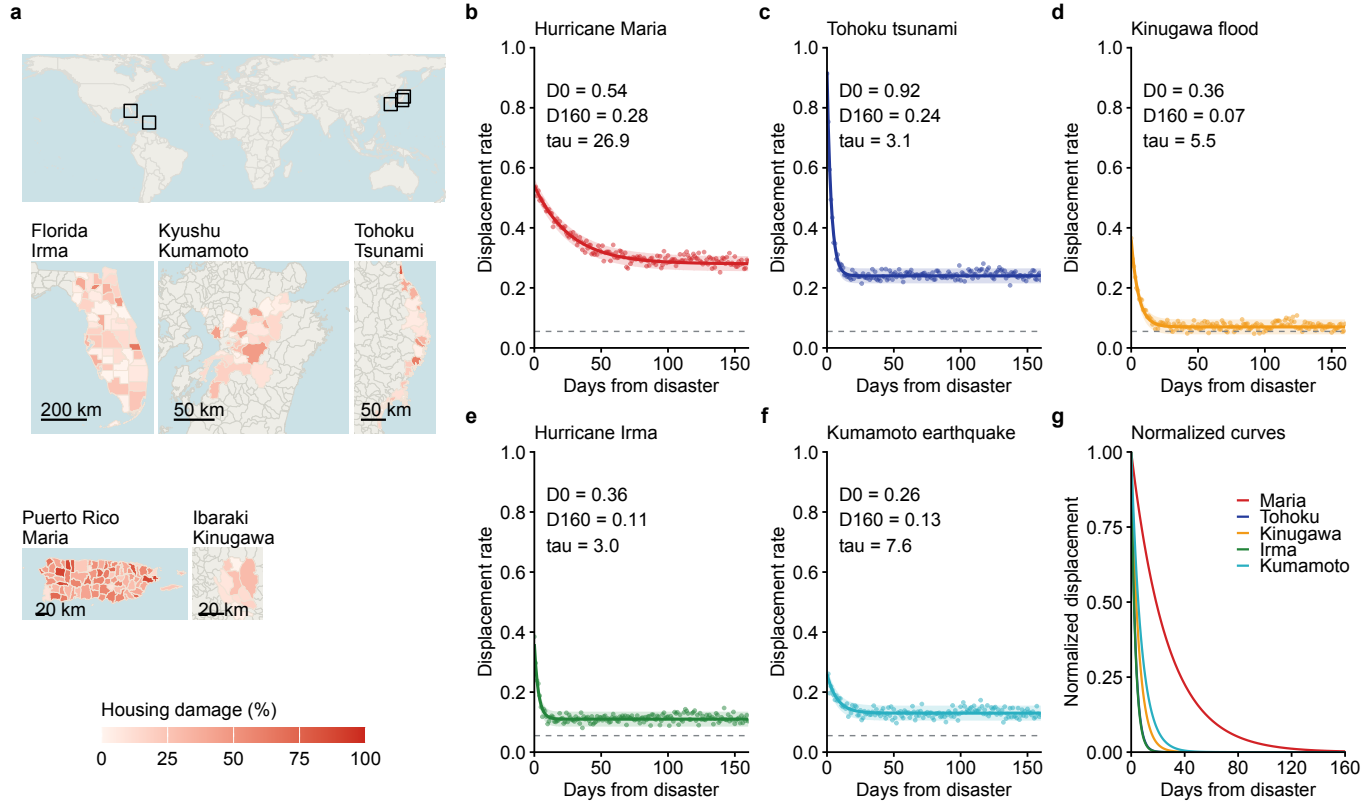

a
Florida
Irma
Kyushu
Kumamoto
Tohoku
Tsunami
200 km
50 km
50 km
Puerto Rico
Maria
Ibaraki
Kinugawa
20 km
20 km
Housing damage (%)
0
25
50
75
100
b
Hurricane Maria
D0 = 0.54
D160 = 0.28
tau = 26.9
c
Tohoku tsunami
D0 = 0.92
D160 = 0.24
tau = 3.1
d
Kinugawa flood
D0 = 0.36
D160 = 0.07
tau = 5.5
e
Hurricane Irma
D0 = 0.36
D160 = 0.11
tau = 3.0
f
Kumamoto earthquake
D0 = 0.26
D160 = 0.13
tau = 7.6
g
Normalized curves
Maria
Tohoku
Kinugawa
Irma
Kumamoto
Displacement rate
Normalized displacement
Days from disaster